A model-based shrinkage target to avoid nonconvergence in small sample SEM

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Abstract

Structural equation modeling is prone to a variety of problems when the sample size is small. One solution that attempts to solve the (nonconvergence) problem of small sample SEM is found in shrinkage estimation, where a weighted average between the sample variance-covariance matrix ($\mathbf{S}$) and a highly structured shrinkage target ($\mathbf{T}$) is calculated to construct an adjusted sample variance-covariance matrix ($\mathbf{S}_a$), which is then used as input for the analysis. Different target candidates have already been put forward in the literature, but in this paper, we propose using a model-based target matrix specifically tailored for structural equation models. Two simulation studies demonstrate the benefit of using a model-based target over other candidate targets in terms of convergence rate, bias and MSE, but also emphasize the importance of constructing an optimal shrinkage intensity.

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Introduction

The problems that arise when SEM is used in settings where the sample size is small have been known for a long time. Indeed, small sample SEM is prone to problems such as extreme parameter values, biased point estimates, negative variances, unreliable p-values, or nonconverged solutions (Bentler & Yuan, 1999; Nevitt & Hancock, 2004). Over the years, a number of proposals have been put forward that attempt to solve the problems of small sample SEM. A first solution that is suggested stems from the Bayesian world where priors are assigned to the various parameters of the model (Song & Lee, 2012; Muthén & Asparouhov, 2012). Bayesian statistics, unlike the frequentist approach, do not rely on asymptotics. For this reason, many problems related to small sample inference are avoided in the Bayesian framework. Although the Bayesian method seems to yield favorable results (Lee & Song, 2004; Muthén & Asparouhov, 2012; Depaoli, 2013; van de Schoot et al., 2017), choosing priors for each and every free parameter in the model is far from trivial. Therefore, care should be taken when choosing those priors (Smid & Rosseel, 2020) and it is often suggested to accompany a Bayesian analysis with a sensitivity analysis where the impact of the priors is investigated (Van Erp, Mulder, Oberski, 2018).

A second solution involves constrained maximum likelihood estimation. In a paper by Ulitzsch et al. (2021) the performance of constrained ML and Bayesian methods using Markov chain Monte Carlo was compared to unconstrained (standard) ML in small sample SEM. Constrained ML tries to prevent inadmissible solutions by restricting the parameter space during estimation. To prevent the estimated parameters from taking on values located at (or on the wrong side of) the boundaries of the admissible parameter space, constrained ML adds (or subtracts) a small value, ϵ, from those boundaries (where ϵ is set at, say, 0.01) and forces the parameters to stay within those boundaries during estimation. As a result, estimated variances for example can never be smaller than ϵ. Their study showed that constrained ML led to much higher convergence rates compared to standard ML. A similar approach was put forward by De

Jonckere & Rosseel (2022). They proposed a method referred to as bounded estimation to specifically avoid the nonconvergence issue in small sample SEM. In bounded estimation, data-driven lower and upper bounds are defined for a subset of the model parameters: the (residual) variances of the observed variables, the (residual) variances of the latent variables and the factor loadings. The bounds are computed prior to estimation. The general idea is that these bounds try to guide the optimizer in the right direction when estimating the parameter values without restricting the admissible parameter space, in contrast to the constrained ML method used in the study of Ulitzsch et al. (2021). By the latter we mean that estimated parameters could still be located at the boundaries. De Jonckere & Rosseel (2022) even allowed for parameters to take on values located slightly outside their admissible bounds, still allowing for, for example, negative variances. Simulation results showed that bounded estimation led to increased rates of convergence as compared to standard ML estimation and this without any (negative) effects on the point estimates for the unbounded parameters (De Jonckere & Rosseel, 2022). The "wide bounds" condition in which the parameter values were allowed to be located outside the admissible bounds, turned out to yield the most favorable results.

A third method tries to prevent the problem of nonconvergence by slightly adjusting or "regularizing" the input variance-covariance matrix. Assuming the data are continuous and complete, the variance covariance matrix is at the core of a SEM analysis. However, due to several factors (such as sampling error and measurement error), the structure of the observed variance-covariance matrix, $\mathbf{S}$, may differ significantly from the one in the population, $\mathbf{\Sigma}$, especially when the sample size is small. If the observed variance-covariance matrix is poorly conditioned, this may lead to poor performance (Yuan & Bentler, 2017). An approach by Arruda & Bentler (2017) adjusts the eigenvalues of ill-conditioned matrices by pushing them towards the geometric mean of the sample eigenvalues. Their simulation studies showed that inputting such a regularized matrix in SEM (for the GLS estimator) outperformed standard GLS and ML at smaller sample sizes (in terms of rejection rates and improper solutions).

Another approach is based on ridging (Hastie,Tibshirani, & Friedman, 2008, p. 61). The general idea of ridging is to add a small (positive) constant, α, to the diagonal of the sample variance-covariance matrix. This is equivalent to: $\mathbf{S} + \alpha\mathbf{I}$, where $\mathbf{I}$ is the identity matrix. Applied to structural equation modeling, studies showed that this type of (co)variance matrix manipulation causes the convergence rate to increase compared to vanilla maximum likelihood without (negatively) affecting the quality of the parameter estimates (Yuan & Chan, 2008; Yuan, Wu & Bentler, 2011; Yuan & Chan, 2016). A similar method was proposed by Ledoit and Wolf (Ledoit & Wolf, 2003, 2004a, 2004b). They suggested modifying $\mathbf{S}$ by means of a weighted average between the sample variance-covariance matrix and a second, highly structured matrix which is often referred to as the shrinkage target ($\mathbf{T}$). This shrinkage target can be, for example, a diagonal (unit variance) matrix, a constant correlation matrix (Ledoit & Wolf, 2003), a diagonal, common variance matrix, and so forth. The method implies regulating both matrices (the sample variance-covariance matrix and the shrinkage target) and is equivalent to $(1 - \lambda)\mathbf{S} + \lambda\mathbf{T}$, where λ is called the shrinkage intensity with $0 \leq \lambda \leq 1$.

Although the aforementioned proposals of Ledoit & Wolf show promising results, they have been developed specifically for stock returns data. Therefore, they do not entirely cover the models that are often fit (using SEM analyses) in the social and behavioral sciences. On the other hand, ridge SEM only slightly modifies the diagonal elements of the sample variance-covariance matrix (i.e. the variances), leaving other elements unaffected. The goal of this paper is to find out if we can do a better job by slightly adjusting the off-diagonal elements within that matrix. We will do so by means of a model-based target which has the sole purpose of adding extra structure to the variance-covariance matrix. This paper will provide a rationale to obtain this model-based target. Two simulation studies are then set up to evaluate the performance of the adjusted variance-covariance matrix (resulting from combining $\mathbf{S}$ with this model-based target) in terms of convergence rate, mean bias and mean squared error (MSE).

The remainder of the paper is organized as follows. In a first section we will briefly elaborate on model estimation in structural equation modeling and the importance of the (sample) variance-covariance matrix. In a second section, the different targets that have already been proposed in the literature will be discussed. In a third section we will introduce and provide a rationale for our model-based target. The next section reports the results of two simulation studies that investigate the performance of the model-based target in comparison with standard ML and other target matrices. We end our paper with a discussion and some ideas for future research.

Model estimation in structural equation modeling

If we assume multivariate normal (and complete) data, $\boldsymbol{\mathcal{Y}} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the mean vector, $\bar{\mathbf{y}}$, and the sample variance-covariance matrix, $\mathbf{S}$, are sufficient statistics. In this paper, we assume that the observed variables are centered so that $\boldsymbol{\mu} = \bar{\boldsymbol{y}} = \mathbf{0}$, making $\mathbf{S}$ our only statistic of interest. If we use $\boldsymbol{\theta}$ to denote the free parameters of our model, then the model-implied variance-covariance matrix can be written as $\boldsymbol{\Sigma}(\boldsymbol{\theta})$. Estimation then proceeds by trying to find those values for $\boldsymbol{\theta}$ that minimize the difference between $\mathbf{S}$ and $\boldsymbol{\Sigma}(\boldsymbol{\theta})$. The difference between both matrices can be expressed by a discrepancy function, $\mathrm{F}(\mathbf{S}, \boldsymbol{\Sigma}(\boldsymbol{\theta}))$. Perhaps the most commonly used discrepancy function in SEM is the normal maximum likelihood discrepancy function:

$$\mathrm{F}_{ML}(\boldsymbol{\theta}) = \log|\boldsymbol{\Sigma}(\boldsymbol{\theta})| - \log|\mathbf{S}| + \mathrm{tr}(\mathbf{S}\boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}) - p, \tag{1}$$

where $|\cdot|$ denotes the determinant of a matrix, $\mathrm{tr}(\cdot)$ the trace of a matrix (i.e. sum of its diagonal elements), and p the number of observed variables. Alternative discrepancy functions, based on the least squares principle, are also available (Browne & Arminger, 1995).

An implicit assumption we make when estimating the parameters in SEM is that $\mathbf{S}$ is a good estimator of the population variance-covariance matrix, $\boldsymbol{\Sigma}$. However, the structure of the sample variance-covariance matrix may (heavily) deviate from the one in the population due to measurement error and sampling error, particularly in the case of small samples. If $\mathbf{S}$ is far away from $\boldsymbol{\Sigma}$, this may lead to estimation problems, as for example nonconvergence. A solution to this problem might be to add additional structure to the (rather unstructured) $\mathbf{S}$. As mentioned in the introduction of this paper, this can be accomplished by means of a highly structured target ($\mathbf{T}$) using: $(1-\lambda)\mathbf{S}+\lambda\mathbf{T}$. In the literature, different proposals for choosing such a target matrix have already been put forward. In the next section of this paper, we will briefly discuss some of these proposals.

Commonly used target matrices

An overview of some commonly used target matrices can be found in Table 1. Our overview is based on a table provided by Schäfer & Strimmer (2005). A first set of commonly used targets each shrink the covariances of $\mathbf{S}$ towards zero and are sparse in terms of (free) parameters . However, they differ in how the diagonal elements (i.e. the variances) are modified. The simplest target matrix is the diagonal, unit variance matrix ($\mathbf{T}_{\mathrm{uv}}$). This matrix adds a (small) constant to the variances and is often used in ridging (i.e. $\mathbf{S}+\alpha\mathbf{I}$). The diagonal, common variance matrix ($\mathbf{T}_{\mathrm{cv}}$), on the other hand, shrinks the variances towards their mean. A third and last target within this set does not impose shrinkage on the variances and is often referred to as a diagonal, unequal variance target ($\mathbf{T}_{\mathrm{unv}}$). Most of these targets were, for example, used in studies reported by Ledoit & Wolf (2003, 2004a). The latter, $\mathbf{T}_{\mathrm{unv}}$, was used in a study by Schäfer & Strimmer (2005). Another shrinkage target as suggested by Ledoit (1995) shrinks the diagonal elements towards the average sample variance and the off-diagonal elements towards the average sample covariance ($\mathbf{T}_{\mathrm{ccv}}$). The last two targets displayed in Table 1 both leave the variances of $\mathbf{S}$ unaffected, but apply shrinkage to the covariances of the matrix. The constant correlation matrix ($\mathbf{T}_{\mathrm{cc}}$) takes the average of the sample correlations and shrinks covariances towards $\bar{\mathrm{r}}\sqrt{\mathrm{s}_{ii}\mathrm{s}_{jj}}$ (Ledoit & Wolf, 2003). When one sets $\bar{\mathrm{r}}=1$, this target switches to the perfect positive correlation target ($\mathbf{T}_{\mathrm{pc}}$). The target assumes perfect correlation between variables and hence shrinks the covariances towards $\sqrt{\mathrm{s}_{ii}\mathrm{s}_{jj}}$.

Studies employing these targets in various fields of research show superior performance over the unadjusted sample variance-covariance matrix. Depending on the study area in which one wishes to apply shrinkage using these targets, not all targets are equally representative of the model one is looking to investigate. As mentioned earlier, the targets used or suggested by Ledoit & Wolf were primarily focused on analyzing stock returns data. The structure of the sample variance-covariance matrix in a SEM analysis may deviate substantially from the target structures we just discussed. In what follows, we provide a rationale for creating a model-based target specifically for analyzing SEM models.

A model-based target matrix for SEM

The general idea is as follows: Given a model, generate a target matrix, $\mathbf{T}_{\text{SEM}}$, that reflects the model. Then, modify the sample variance-covariance matrix, $\mathbf{S}$, in the following way:

$$\mathbf{S}_a = (1-\lambda) \times \mathbf{S} + \lambda \times \mathbf{T}_{\text{SEM}}, \tag{2}$$

where $0 \leq \lambda \leq 1$, and use $\mathbf{S}_a$ instead of $\mathbf{S}$ as input for model estimation.

The model-based target is constructed by expressing the target matrix (in correlation metric) as a CFA model: $\mathbf{T}^*_{\text{SEM}} = \mathbf{\Lambda}^*\mathbf{\Psi}^*\mathbf{\Lambda}^* + \mathbf{\Theta}^*$, where the * superscript reflects standardized matrices. Before we can proceed in constructing $\mathbf{S}_a$, we will first need to rescale $\mathbf{T}^*_{\text{SEM}}$ so that the diagonal of $\mathbf{T}_{\text{SEM}}$ coincides with the diagonal of $\mathbf{S}$. This is achieved by $\mathbf{T}_{\text{SEM}} = \mathbf{D}\mathbf{T}^*_{\text{SEM}}\mathbf{D}$, where $\mathbf{D} = \text{diag}(\sqrt{\text{s}_{11}}, \sqrt{\text{s}_{22}}, \ldots \sqrt{\text{s}_{ii}})$. Once this is done, we construct $\mathbf{S}_a$ using (2). The rationale behind constructing $\mathbf{T}^*_{\text{SEM}}$ will be explained by focusing on the measurement part ($\mathbf{\Lambda}^*$ and $\mathbf{\Theta}^*$) and the structural part ($\mathbf{\Psi}^*$) of a SEM model separately (Figure 1).

The measurement part

The measurement part of a SEM model includes the (standardized) factor loadings of the observed variables and their error variances. The idea is that we fill in ideal parameter values that will lead us to an ideal $\mathbf{\Sigma}^*(\boldsymbol{\theta})$. For the factor loadings ($\mathbf{\Lambda}^*$), we choose to set all loadings equal to (the arbitrary value of) 0.7. By setting them all equal to 0.7, we assume that each indicator is equally good at measuring its corresponding latent variable. Because ideally, we like to have strong measurement models, we additionally set the reliability of our indicators equal to 0.8. The (diagonal) elements of $\mathbf{\Theta}^*$ are then chosen in such a way so that the indicator reliability is exactly 0.8. Additionaly, we assume that $\mathbf{\Theta}^*$ is a diagonal matrix (i.e. no correlated error terms), regardless of what the initial model under consideration looks like.

The structural part

Before we proceed in calculating $\mathbf{T}^*_{\text{SEM}}$, we first need to define $\mathbf{\Psi}^*$. We will assume a saturated model for the structural part, so we only need to specify the correlations among the latent variables. We consider several options for obtaining these correlations in $\mathbf{\Psi}^*$. A first option implies inserting user-defined correlations. It may very well be that the researcher possesses prior knowledge about the model-implied correlations between the latent factors (e.g. based on previous research) and this information can be used by directly filling in these correlations in constructing $\mathbf{\Psi}^*$. A second option is to specify a single correlation for all possible (latent) correlations in the model. Depending on prior expectations, the user may prefer a stronger or weaker correlation. If the user also has strong prior knowledge about the sign of the correlations, this can also be included. If one does not wish to make any assumption about the direction nor about the strength of the linear relationship between the latent variables, one could choose to set the correlation between the latent variables to zero. This leads to the third option where $\mathbf{\Psi}^*$ is a diagonal matrix and, thus, the inter-factor blocks of $\mathbf{S}$ are shrunken towards zero. This will typically result in underestimation, but in turn will result in lower variability. The focus of this option is on adding extra structure to the measurement part of the model only, not on what happens between latent factors. A variation of the latter option leads us to a fourth option in constructing $\mathbf{\Psi}^*$. In this fourth option, not only the variances but also the off-diagonal elements of the inter-factor blocks are left unaffected. This implies setting $\mathrm{t}_{ij} = \mathrm{s}_{ij}$ (or $\mathrm{t}^*_{ij} = \mathrm{s}^*_{ij}$) for the inter-factor blocks. Table 2 gives an overview of the different $\mathbf{T}^*_{\text{SEM}}$ matrices.

Example

To illustrate the use of $\mathbf{T}_{\text{SEM}}$, consider the example in Appendix A. Here, we consider a two-factor model with three indicators for each latent variable and $\beta = 0.25$ (Figure 2). Given a sample size of 20, we fit a model using the `lavaan` package (version 0.6.10) in R (Rosseel, 2012). Unfortunately, it turns out the model did not reach convergence (Appendix B). In a next step, we adjust the sample variance-covariance matrix by using (2). For λ, we choose a value of 0.2. For this example, we also fit the model using all four options for $\mathbf{\Psi}^*$. The resulting adjusted variance-covariance matrices for all options of $\mathbf{T}_{\text{SEM}}$ are depicted in Table 3. When fitting the model using $\mathbf{S}_a^{1-3}$, the reader can verify that the model converged for all options of $\mathbf{T}_{\text{SEM}}$. For $\mathbf{S}_a^4$, however, we had to increase λ to 0.25 until convergence was reached.

Choosing the shrinkage intensity

The parameter λ in (2) regulates the amount of influence both the sample variance-covariance matrix and the target matrix have in calculating $\mathbf{S}_a$. The parameter λ is often referred to as the shrinkage intensity. How this value for λ should be obtained can depend on many different factors. In the aforementioned example, we chose to set λ equal to 0.2. However, it might well be that if we chose the value for λ to be 0.1, convergence is not reached. In fact, for this specific example, convergence is not reached until λ equals (about) 0.2 and even 0.25 for $\mathbf{S}_a^4$. In a situation where $\mathbf{S}$ is a good representation of $\mathbf{\Sigma}$, the value for λ should preferably be smaller. In doing so, the weight that is given to the sample variance-covariance matrix when calculating $\mathbf{S}_a$ is larger. In a situation where $\mathbf{S}$ is not a good representation of $\mathbf{\Sigma}$, we would prefer to give more structure to the sample variance-covariance matrix by choosing a larger value for λ. There are some proposals in the literature for finding the optimal value for λ. One approach is cross-validation (Browne, 2000), where the sample is divided into a training dataset and a test dataset. The "optimal" value for λ is then found by fitting different models for a set of λ values on the training dataset, and re-fitting (using the estimated parameters in the previous step as fixed values) the model on the test dataset to finally pick the model that resulted in the best fit on the test dataset (Jacobucci, Grimm, & McArdle, 2016). The value for λ used to obtain this best model fit is then put forward as the most suitable value. In small samples, however, the initial sample is already (very) small, and splitting it further into a training dataset and test dataset may not be ideal. In addition, our focus is not in obtaining the best fit. Instead, we merely wish to obtain convergence. Therefore, a simple strategy may be to look for the smallest value of λ that causes the model to converge. We will investigate this option in study 1.

Another option to retrieve an optimal value for λ is proposed by Ledoit & Wolf (2003, 2004a, 2004b) who suggested calculating the shrinkage intensity by minimizing a loss-function. They define the optimal shrinkage intensity as δ/N with

$$\delta = \frac{\pi - \rho}{\gamma}, \tag{3}$$

where π reflects the sum of asymptotic variances of the elements in $\mathbf{S}$ scaled by $\sqrt{N}$ (N = number of observations), ρ the sum of asymptotic covariances of the elements in $\mathbf{S}$ scaled by $\sqrt{N}$, and γ measuring the misspecification of the model that is used for $\mathbf{T}$ (Ledoit & Wolf, 2003). In the second simulation study, we will use (3) to obtain λ.

Simulation studies

Two simulation studies [1] were set up to assess the impact of using a weighted average of a highly structured model-based target $\mathbf{T}_{\text{SEM}}$ and the sample variance-covariance matrix $\mathbf{S}$ on the convergence rate in small sample SEM. The goal of the first study is to compare the effect of utilizing different shrinkage targets in constructing the adjusted variance-covariance matrix. In the second study, we evaluated the effect of a variety of values for β and number of observed variables (P). The value for the shrinkage intensity was obtained automatically. Both simulation studies were implemented using R, version 4.1.2 (R Core Team, 2021) and `lavaan` 0.6.10 (Rosseel, 2012).

[1] The code that was used for these two studies can be found using the following link: https://osf.io/8qp3f/

Study 1

In the first study, two different data-generating models are considered (see Figure 2). The population value for β is either 0.1 or 0.3. Both models have six observed variables (P = 6). Six sample sizes are considered: 10, 20, 30, 40, 50, 100. For each combination of N and β, a thousand random samples of size N are generated (drawn from $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$). Each replication implies fitting six SEM models: a model using standard maximum likelihood (ML), a model using an adjusted variance-covariance matrix where a user-defined correlation is specified for the model-based target (i.e., $\rho = 0.2$; MB1), a model using an adjusted variance-covariance matrix where no assumption about the strength nor about the direction of the correlation is made for the model-based target (i.e., $\rho = 0$; MB2), a model using an adjusted variance-covariance matrix with an identity target (ID), a model using an adjusted-variance covariance matrix with a common covariance target (CV), and lastly, a model using an adjusted variance-covariance matrix with a constant correlation target (see Table 1; CC). The value for λ is chosen sequentially out of 51 different candidate values, with $0 \leq \lambda \leq 1$. From 0.1 to 0.95 the gap between two subsequent numbers equals 0.02. For every value of $\lambda < 0.1$ or > 0.95, smaller gaps are used (e.g., 0.01, 0.001, 0.0001). The first value for which the model converges is then chosen as the most appropriate value.

Table 4 depicts the amount of nonconverged cases for standard ML (ML_1 and ML_2) and different $\mathbf{S}_a$'s. It appears that the number of nonconverged cases differs per target, with zero nonconverged cases only for the MB1 and MB2 targets. Also for the second model, convergence is (almost) always reached for $\mathbf{S}_a$ using the MB1, MB2, or CC target. It seems that for higher values of β, the number of nonconverged cases decreases for the ID, CV and CC targets as compared to model 1. The values between brackets in Table 4 represent the mean value for the shrinkage intensity for the different choices of $\mathbf{S}_a$. As an example, the distribution of λ at a sample size of 30 for model 1 and model 2 can be found in Figure 5 and Figure 6, respectively.

The top part of Table 5 shows the mean bias ($\times 100$) for different target matrices and standard ML. In general, bias decreases as sample size increases. The same holds for a larger value of β. For both models it can be seen that in most cases, bias for the target matrices is smaller than for standard ML. This pattern is not observed for the constant correlation target. For model 1, differences in bias between the target matrices are not apparent. Only the constant correlation target seems to yield the least favorable results. This pattern is retained for model two. Note that bias values are based on all cases that converged, also those cases where $\lambda = 0$. For example, the bias value for the MB1 target at sample size 20 for a value of $\beta = 0.1$ is based on 1000 cases (no nonconverged cases for this target at sample size 20), whereas the bias value for the ID target at sample size 20 is based on 965 cases (see Table 4).

The bottom part of Table 5 depicts the MSE ($\times 100$). In general, as sample size increases, MSE values decrease. Looking at the smaller sample sizes, MSE values are smallest for the model-based targets for both model 1 and model 2. In some conditions, Table 5 shows the rather undesirable result that MSE values for the targets are higher than is observed for standard ML. Note, however, that we did not include values for λ that try to minimize the MSE, but rather used values that had the sole purpose of causing the model to reach convergence. It may well be that the value for λ that causes the model to converge, is not identical to the value that leads towards smaller MSE values. In addition, just like for the bias, the MSE values for ML are only based on those cases that converged with $\lambda = 0$.

Overall, it seems that more models reach convergence after an adjusted variance-covariance matrix is constructed using the MB1, MB2 or CC target. Only for MB1 and MB2, this coincides with lower biases, lower MSE values, and lower mean values for the shrinkage intensity, λ. This indicates that, on average, less 'adjustment' of the sample variance-covariance matrix is needed using this type of shrinkage target.

Study 2

In the second study, both the values for β and P are varied (see Figures 2 - 4). Values for β are either 0, 0.25, 1 or 2. The number of observed variables (P) is either 6, 14 or 18. As before, six sample sizes are considered: 10, 20, 30, 40, 50, 100. In this second study, a more sophisticated value for λ is obtained and used in constructing the adjusted (co)variance matrices. At each replication, three SEM models are fit: a model using standard maximum likelihood estimation (ML), a model using an adjusted (co)variance matrix with a model-based shrinkage target and a value for λ as in (3) (MB_l), and lastly, a model using an adjusted (co)variance matrix with a constant correlation target and λ as proposed by Ledoit & Wolf (3; LW).

In Table 6 the reader may find the number of nonconverged solutions for different values of β, sample size, and number of observed variables (P). In general, results show that the larger the value for β, the smaller the number of nonconverged cases. Additionally, for increasing N and increasing P, a decrease in the number of nonconverged solutions is found. Overall, it seems that standard ML results in the worst performance in terms of nonconvergence, especially at lower sample sizes. Moving forward, it seems that (clear) differences between both shrinkage methods only appear for P = 6. These differences seem to vanish as P increases. For P $\geq$ 14 and N $\geq$ 40, the amount of nonconverged cases drops to zero for both adjusted variance-covariance matrices. For $\beta = 0$, the MB_l method results in less nonconverged cases. This observation is less prominent for $\beta = 0.25$ and even reversed for $\beta \geq 1$.

Bias ($\times$100) for β is depicted in Table 7. As expected, results show that with increasing sample size, bias decreases. For smaller values of β, the MB_l method results in smaller (mean) bias. However, for $\beta \geq 1$, the LW method elicits smaller bias. At all conditions, it seems that the least favorable shrinkage method results in bias values that are larger than what is observed for standard ML. Only for standard ML, bias drops as P increases.

Table 8 gives an overview of the MSE ($\times$100) values for the different methods and conditions. In general, MSE values decrease as sample size and the number of observed variables increase. One exception is found for the ML method at P = 18, N = 30 and $\beta = 2$. At all conditions, a similar pattern as in Table 7 is observed, where smaller MSE values are found for the MB_l method at smaller β, and smaller MSE values for the LW method at larger β. Differences between all methods vanish at N = 100.

In general, it seems that the model-based target performs best at smaller β. This is not surprising as ρ was set to 0 when constructing the model-based target. The constant correlation target by Ledoit & Wolf calculates $\bar{r}\sqrt{s_{ii}s_{jj}}$ and, thus, takes into account the average correlation (of the observed variables) for each dataset separately, while the model-based target keeps the value for ρ (correlation latent variables) fixed. However, it seems that the constant correlation target endures more trouble when the value for β is close to zero. Note that, despite the fact that the LW performs best at higher β, even if we fix ρ at 0 for the model-based target, results still show lower MSE values as compared to standard ML.

Combining results from study 1 and study 2, it seems that the shrinkage intensity used in the second study makes up for the higher bias and MSE values that accompany the constant correlation target in study 1 (compared to bias and MSE values for the model-based target). It may well be that a more appropriate shrinkage target can be constructed for the model-based target that may lead to better performance.

Discussion

Nonconvergence is one of many problems that may arise when SEM is used in settings where the sample size is small. In the literature, some proposals have been made that try to tackle (nonconvergence) problems related to small sample SEM. Taking a weighted average of a highly structured target and the sample variance-covariance matrix and using this adjusted (co)variance matrix as input when estimating a model, turns out te be an effective method to overcome (some of) those problems. An important element of such an adjusted (co)variance matrix is the shrinkage target. With the nonconvergence issue as main interest, this paper provided a rationale for constructing such a model-based shrinkage target, incorporating characteristics of SEM models.

In line with previous research, we were able to find better performance in terms of convergence, bias and MSE for adjusted variance-covariance matrices as compared to standard maximum likelihood. Applying a model-based target not only significantly decreases the amount of nonconverged cases, it also requires less adjustment to the sample covariance matrix as compared to other target matrices, especially at smaller sample sizes, smaller β and small P. However, when the Ledoit & Wolf method is used to set the shrinkage intensity and we compare performance of the model-based target to the constant correlation target differences are not remarkably better, but also not worse, at all conditions.

The rationale behind the adjusted variance-covariance matrix using a model-based target has nice intuitive features that strongly relate to the Bayesian framework. In our approach, prior information is automatically translated to a model-based target matrix which is then used to modify the sample input variance-covariance matrix. This is in contrast with the Bayesian method, where priors are needed for each and every parameter in the model, which at first sight might discourage researchers that are not familiar with Bayesian statistics. Clearly, the more accurate the prior knowledge of the researcher is, the better the model-based target will perform. The same holds for the use of (informative) priors in the Bayesian framework.

In our simulation study, we did not thoroughly investigate the vital role of (accurate) prior knowledge on the structural part of the model-based target. The impact prior knowledge may have on the performance of the model-based target should be subject to future research. Additionally, we were not able to show superior performance of our model-based target in all conditions, despite the fact that it contains more (SEM) model specific characteristics than for example the constant correlation target. As was already mentioned in the results section of this paper, this may be due to the approach that was used to obtain the shrinkage intensity. In future work we could try to construct a more sophisticated, a priori formula for the model-based shrinkage target. A third shortcoming is that we only considered fairly simple and correctly specified models. Future research could examine the performance of the model-based targets in more complex models and/or when a certain amount of misspecification is present. As a last remark, it may be interesting to consider a combination of the bounded estimation approach by De Jonckere & Rosseel (2022) and the model-based target. This approach could use the model-based target in case the model does not reach convergence when, for example, only lower bounds on the (observed and latent) variances are imposed. How this may affect bias and MSE values can be explored in future research.

To conclude, a (SEM) model-based shrinkage target effectively decreases the amount of nonconverged cases as compared to other target matrices for small sample SEM. Despite the fact that we were only able to show superior performance at smaller sample sizes, smaller β and smaller number of observed variables, we believe that the rationale behind the use of a model-based target (i.e., taking into account specific SEM model characteristics), is more consistent with research using SEM analyses. Nonetheless, some questions (e.g., how to find the optimal shrinkage intensity) remain and more research on this topic is needed.

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Target Matrices

Unit variance(I)	**Common variance**	**Unequal variance**
$\mathbf{T}_{\mathrm{uv}} = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & 0 & 1 \end{bmatrix}$	$\mathbf{T}_{\mathrm{cv}} = \begin{bmatrix} \mathrm{v} & & \\ 0 & \mathrm{v} & \\ 0 & 0 & \mathrm{v} \end{bmatrix}$	$\mathbf{T}_{\mathrm{unv}} = \begin{bmatrix} \mathrm{s}_{11} & & \\ 0 & \mathrm{s}_{22} & \\ 0 & 0 & \mathrm{s}_{33} \end{bmatrix}$
Common (co)variance	**Constant correlation**	**Perfect + correlation**
$\mathbf{T}_{\mathrm{ccv}} = \begin{bmatrix} \mathrm{v} & & \\ \mathrm{c} & \mathrm{v} & \\ \mathrm{c} & \mathrm{c} & \mathrm{v} \end{bmatrix}$	$\mathbf{T}_{\mathrm{cc}} = \begin{bmatrix} \mathrm{s}_{11} & & \\ \bar{\mathrm{r}}\sqrt{\mathrm{s}_{11}\mathrm{s}_{22}} & \mathrm{s}_{22} & \\ \bar{\mathrm{r}}\sqrt{\mathrm{s}_{11}\mathrm{s}_{33}} & \bar{\mathrm{r}}\sqrt{\mathrm{s}_{22}\mathrm{s}_{33}} & \mathrm{s}_{33} \end{bmatrix}$	$\mathbf{T}_{\mathrm{pc}} = \begin{bmatrix} \mathrm{s}_{11} & & \\ \sqrt{\mathrm{s}_{11}\mathrm{s}_{22}} & \mathrm{s}_{22} & \\ \sqrt{\mathrm{s}_{11}\mathrm{s}_{33}} & \sqrt{\mathrm{s}_{22}\mathrm{s}_{33}} & \mathrm{s}_{33} \end{bmatrix}$

Table 1

Overview of six target matrices with $v = \frac{1}{p}\sum_{i=1}^{p} s_{ii}$, $c = \frac{1}{p^*}\sum_{i=1}^{p-1}\sum_{j=(i+1)}^{p} s_{ij}$, *and* $\bar{r} = \frac{1}{p^*}\sum_{i=1}^{p-1}\sum_{j=(i+1)}^{p} r_{ij}$, *where* $p^* = p(p-1)/2$.

Option 1

$$\mathbf{T}^*_{\text{sem}} = \begin{bmatrix} 1 & & & & & \\ .80 & 1 & & & & \\ .80 & .80 & 1 & & & \\ .32 & .32 & .32 & 1 & & \\ .32 & .32 & .32 & .80 & 1 & \\ .32 & .32 & .32 & .80 & .80 & 1 \end{bmatrix}$$

Option 2

$$\mathbf{T}^*_{\text{sem}} = \begin{bmatrix} 1 & & & & & \\ .80 & 1 & & & & \\ .80 & .80 & 1 & & & \\ -.16 & -.16 & -.16 & 1 & & \\ -.16 & -.16 & -.16 & .80 & 1 & \\ -.16 & -.16 & -.16 & .80 & .80 & 1 \end{bmatrix}$$

Option 3

$$\mathbf{T}^*_{\text{sem}} = \begin{bmatrix} 1 & & & & & \\ .80 & 1 & & & & \\ .80 & .80 & 1 & & & \\ 0 & 0 & 0 & 1 & & \\ 0 & 0 & 0 & .80 & 1 & \\ 0 & 0 & 0 & .80 & .80 & 1 \end{bmatrix}$$

Option 4

$$\mathbf{T}^*_{\text{sem}} = \begin{bmatrix} 1 & & & & & \\ .80 & 1 & & & & \\ .80 & .80 & 1 & & & \\ \mathrm{r}_{14} & \mathrm{r}_{24} & \mathrm{r}_{34} & 1 & & \\ \mathrm{r}_{15} & \mathrm{r}_{25} & \mathrm{r}_{35} & .80 & 1 & \\ \mathrm{r}_{16} & \mathrm{r}_{26} & \mathrm{r}_{36} & .80 & .80 & 1 \end{bmatrix}$$

Table 2

The different (standardized) model-based targets $\boldsymbol{T}^*_{sem}$ for different options of $\boldsymbol{\Psi}^*$ using the population model in Figure 2. The first option implies a user-defined correlation matrix where $\rho_{ij} = .4$. Option two and three have $\rho_{ij} = -.2$ and 0, respectively. The fourth option leaves the covariances (correlations) between indicators unaffected. Note that data is necessary for obtaining $\boldsymbol{T}^*_{sem}$ option 4.

$$\mathbf{S}_a^1 = \begin{bmatrix} 1.65 & & & & & \\ 1.13 & 1.92 & & & & \\ 0.64 & 0.78 & 0.72 & & & \\ 0.32 & 0.15 & 0.04 & 1.26 & & \\ 0.31 & 0.33 & 0.18 & 0.68 & 0.87 & \\ 0.27 & 0.27 & 0.25 & 0.54 & 0.41 & 1.00 \end{bmatrix} \quad \mathbf{S}_a^2 = \begin{bmatrix} 1.65 & & & & & \\ 1.13 & 1.92 & & & & \\ 0.64 & 0.78 & 0.72 & & & \\ 0.21 & 0.02 & -0.03 & 1.26 & & \\ 0.22 & 0.23 & 0.12 & 0.68 & 0.87 & \\ 0.17 & 0.16 & 0.18 & 0.54 & 0.41 & 1.00 \end{bmatrix}$$

$$\mathbf{S}_a^3 = \begin{bmatrix} 1.65 & & & & & \\ 1.13 & 1.92 & & & & \\ 0.64 & 0.78 & 0.72 & & & \\ 0.09 & -0.10 & -0.11 & 1.26 & & \\ 0.12 & 0.12 & 0.06 & 0.68 & 0.87 & \\ 0.07 & 0.05 & 0.12 & 0.54 & 0.41 & 1.00 \end{bmatrix} \quad \mathbf{S}_a^4 = \begin{bmatrix} 1.65 & & & & & \\ 1.13 & 1.92 & & & & \\ 0.64 & 0.78 & 0.72 & & & \\ 0.18 & -0.20 & -0.22 & 1.26 & & \\ 0.24 & 0.25 & 0.11 & 0.68 & 0.87 & \\ 0.13 & 0.10 & 0.23 & 0.54 & 0.41 & 1.00 \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} 1.65 & & & & & \\ 0.84 & 1.92 & & & & \\ 0.42 & 0.62 & 0.72 & & & \\ 0.18 & -0.20 & -0.22 & 1.26 & & \\ 0.24 & 0.25 & 0.11 & 0.53 & 0.87 & \\ 0.13 & 0.10 & 0.23 & 0.18 & 0.08 & 1.00 \end{bmatrix}$$

Table 3

Adjusted ($\boldsymbol{S}_a^1 - \boldsymbol{S}_a^4$) and sample ($\boldsymbol{S}$) variance covariance matrices with $\boldsymbol{S}_a^1$ using $\boldsymbol{T}_{SEM}$ where $\boldsymbol{\Psi}^*$ = option 1, $\boldsymbol{S}_a^2$ using $\boldsymbol{T}_{SEM}$ where $\boldsymbol{\Psi}^*$ = option 2, $\boldsymbol{S}_a^3$ using $\boldsymbol{T}_{SEM}$ where $\boldsymbol{\Psi}^*$ = option 3, and $\boldsymbol{S}_a^4$ using $\boldsymbol{T}_{SEM}$ where $t_{ij} = s_{ij}$ ($\boldsymbol{\Psi}^*$ option 4).

		$\beta = 0.1$						$\beta = 0.3$				
N	ML_1	MB1	MB2	ID	CV	CC	ML_2	MB1	MB2	ID	CV	CC
						Number of nonconverged solutions						
10	485	0 (0.21)	0 (0.22)	48 (0.51)	29 (0.49)	1 (0.33)	463	0 (0.22)	0 (0.23)	42 (0.53)	40 (0.49)	1 (0.35)
20	335	0 (0.15)	0 (0.15)	35 (0.63)	31 (0.59)	0 (0.32)	283	0 (0.16)	0 (0.16)	17 (0.64)	18 (0.60)	0 (0.31)
30	176	0 (0.12)	0 (0.12)	21 (0.66)	23 (0.63)	1 (0.30)	132	0 (0.11)	0 (0.12)	12 (0.63)	14 (0.58)	0 (0.26)
40	131	0 (0.09)	0 (0.09)	15 (0.74)	17 (0.69)	0 (0.26)	86	0 (0.10)	0 (0.10)	6 (0.65)	10 (0.59)	0 (0.22)
50	71	0 (0.09)	0 (0.09)	9 (0.71)	13 (0.66)	0 (0.24)	42	0 (0.07)	0 (0.07)	6 (0.66)	5 (0.64)	0 (0.17)
100	8	0 (0.05)	0 (0.05)	3 (0.75)	2 (0.83)	0 (0.16)	3	0 (0.05)	0 (0.05)	0 (0.74)	1 (0.57)	0 (0.20)
Total		0	0	131	115	2		0	0	83	88	1

Table 4

Number of nonconverged solutions for standard ML (ML_1 and ML_2) and number of nonconverged solutions for choices different $\boldsymbol{S}_a$'s at different sample sizes (1000 replications per sample size). A case was labeled as converged if the optimizer claimed convergence and if all standard errors could be computed. Between brackets the average value for the shrinkage intensity (converged solutions only) are found.

	$\beta = 0.1$						$\beta = 0.3$					
N	MB1	MB2	ID	CV	CC	ML	MB1	MB2	ID	CV	CC	ML
mean bias												
10	-3	-2	5	6	38	-4	11	5	2	7	47	22
20	7	6	0	2	19	13	10	10	18	17	26	13
30	1	1	2	3	6	3	8	8	10	10	14	13
40	2	0	3	4	5	1	4	3	6	6	7	7
50	0	0	-4	-3	2	1	3	3	3	3	4	4
100	0	0	0	0	0	0	0	0	0	0	0	0
MSE												
10	1325	1062	2147	2908	2567	1325	857	430	2578	2434	2195	275
20	199	195	941	1153	259	288	187	180	1078	994	400	88
30	42	41	42	43	42	45	166	150	153	153	185	171
40	87	23	106	131	75	14	21	21	36	37	26	21
50	19	19	174	150	23	20	39	39	39	39	39	40
100	3	3	3	3	3	3	4	4	4	4	4	4

Table 5

Mean bias (×100) and MSE (×100) for different population values of β and sample sizes. The target matrices of interest are a model-based target with user-defined correlation matrix (MB1, with $\rho = 0.2$), a model-based target where no assumption about the strength nor about the direction of the (latent) correlations is made (MB2, $\rho = 0$), an identity target (ID), a common (co)variance target (CV) and a constant correlation target (CC). Performance of the adjusted (co)variance matrices using these targets is compared to standard ML (converged solutions only). Note that the number of replications on which the results are based, differs per cel (see Table 4).

	$\beta = 0$			$\beta = 0.25$			$\beta = 1$			$\beta = 2$		
N	MB_l	LW	ML	MB_l	LW	ML	MB_l	LW	ML	MB_l	LW	ML
P = 6												
10	105	111	514	86	69	482	45	22	274	19	13	155
20	44	66	329	41	36	274	16	6	86	2	2	30
30	22	30	190	15	18	155	2	1	18	0	0	3
40	12	29	136	9	6	92	1	1	7	0	1	2
50	10	16	61	2	4	51	0	0	0	0	0	0
100	0	0	7	0	0	1	0	0	0	0	0	0
P = 14												
20	5	1	146	2	0	127	0	0	54	0	0	31
30	2	1	36	1	1	42	0	0	9	0	0	2
40	0	0	15	0	0	5	0	0	0	0	0	0
50	0	0	2	1	0	4	0	0	0	0	0	0
100	0	0	0	0	0	0	0	0	0	0	0	0
P = 18												
20	2	1	184	3	2	167	0	0	61	0	0	29
30	2	1	53	1	0	42	0	0	12	0	0	3
40	0	0	10	0	0	10	0	0	1	0	0	1
50	0	0	4	0	0	2	0	0	0	0	0	0
100	0	0	0	0	0	0	0	0	0	0	0	0

Table 6

Number of nonconverged solutions for an adjusted variance-covariance matrix with λ as in (3), MB_{lw}), a shrinkage estimator proposed by Ledoit & Wolf using a constant correlation matrix (LW), and standard maximum likelihood (ML) for different population values of β.

	$\beta = 0$			$\beta = 0.25$			$\beta = 1$			$\beta = 2$		
N	MB_l	LW	ML	MB_l	LW	ML	MB_l	LW	ML	MB_l	LW	ML
						P = 6						
10	0	36	9	-14	49	28	-46	28	34	-98	11	97
20	0	41	2	-13	48	9	-43	23	22	-74	-15	38
30	-1	36	0	-10	47	7	-32	21	17	-53	-8	24
40	1	30	2	-9	37	5	-26	20	12	-42	-9	16
50	1	24	1	-8	32	3	-23	18	4	-35	-8	9
100	0	12	0	-5	18	0	-13	12	1	-20	-6	3
						P = 14						
20	1	49	-2	-12	49	2	-43	23	13	-92	-9	21
30	0	40	1	-9	43	2	-35	21	8	-71	-4	25
40	0	34	0	-7	39	2	-28	18	5	-59	-4	9
50	-1	26	-2	-6	34	1	-25	16	2	-51	-5	5
100	0	16	0	-2	24	3	-12	12	3	-28	-1	4
						P = 18						
20	2	46	3	-12	47	0	-41	25	20	-92	2	34
30	0	43	-1	-9	46	1	-34	19	7	-74	-2	28
40	1	38	1	-7	40	1	-27	18	5	-61	0	14
50	-2	29	-2	-5	37	2	-24	15	3	-53	-3	6
100	0	18	0	-2	24	2	-12	11	3	-29	1	6

Table 7

Bias (×100) for $\hat{\beta}$ using adjusted variance-covariance matrix with λ as proposed by Ledoit & Wolf (MB_l), a shrinkage estimator proposed by Ledoit & Wolf using a constant correlation matrix (LW), and standard maximum likelihood (ML) for different population values of β.

	$\beta = 0$			$\beta = 0.25$			$\beta = 1$			$\beta = 2$		
N	MB_l	LW	ML	MB_l	LW	ML	MB_l	LW	ML	MB_l	LW	ML
						P = 6						
10	8	1027	559	14	372	472	272	61	896	157	1396	2473
20	5	89	123	8	58	109	32	41	156	91	31	587
30	19	48	31	6	132	32	23	15	141	54	89	308
40	4	27	16	5	28	29	17	10	87	44	18	72
50	3	18	9	4	22	10	13	8	16	31	15	42
100	2	6	3	2	8	4	7	4	7	13	8	13
						P = 14						
20	6	60	99	8	47	37	26	17	267	102	31	209
30	4	34	19	5	31	26	19	12	49	64	24	181
40	2	23	7	3	24	10	13	9	16	46	17	44
50	3	16	6	3	18	6	10	7	10	35	13	23
100	1	6	2	2	9	3	5	5	5	13	7	9
						P = 18						
20	5	103	47	6	123	59	25	21	137	98	43	263
30	4	36	17	5	37	18	19	12	62	66	24	1137
40	3	26	10	3	24	7	13	10	17	46	17	49
50	3	18	6	3	20	6	10	8	11	36	14	22
100	2	7	2	1	9	2	4	4	4	14	7	10

Table 8

Mean squared error (×100) for $\hat{\beta}$ using the adjusted variance-covariance matrix with λ as proposed by Ledoit & Wolf (MB_l), a shrinkage estimator proposed by Ledoit & Wolf using a constant correlation matrix (LW), and standard maximum likelihood (ML) for different population values of β.

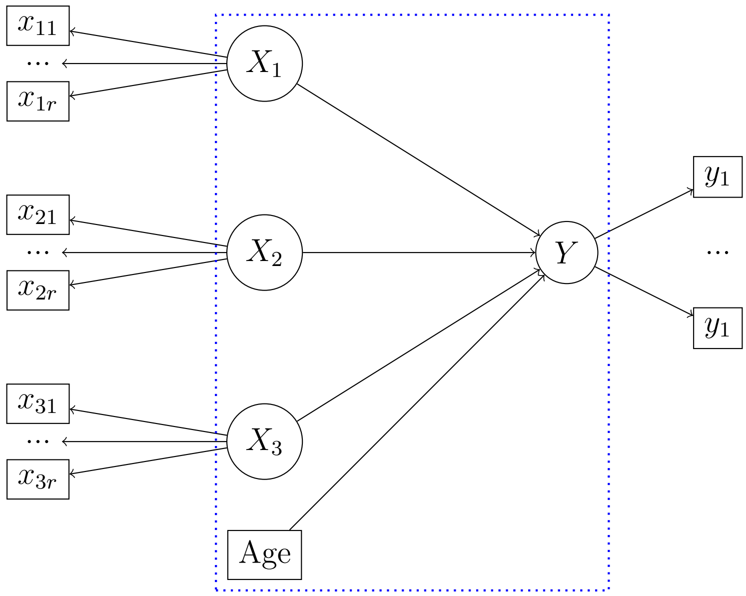

Figure 1. Structural and measurement part of a structural equation model.

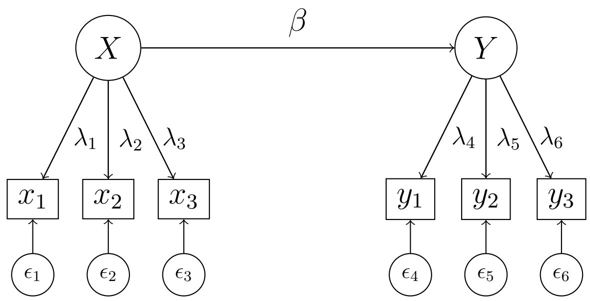

Figure 2. A simple model where the latent variable Y is regressed on the latent variable X. Each latent variable is measured by three observed indicators. The population values are as follows: $\lambda_1 = \lambda_4 = 1$, $\lambda_2 = \lambda_5 = 0.8$, $\lambda_3 = \lambda_6 = 0.6$, all (residual) variances are set to one, and the regression parameter (β) is set to either 0.1 or 0.3 for study 1, and set to 0, 0.25, 1 or 2 for study 2.

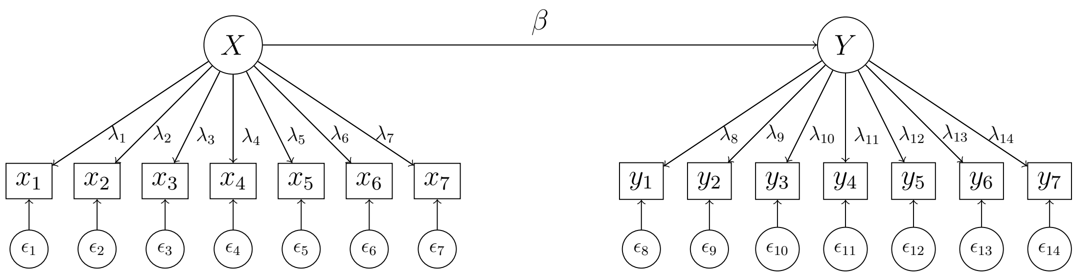

Figure 3. A simple model where the latent variable Y is regressed on the latent variable X. Each latent variable is measured by seven observed indicators. The population values are as follows: $\lambda_1 = \lambda_8 = 1$, $\lambda_2 = \lambda_9 = 0.8$, $\lambda_3 = \lambda_{10} = 0.7$, $\lambda_4 = \lambda_{11} = 0.6$, $\lambda_5 = \lambda_{12} = 0.5$, $\lambda_6 = \lambda_{13} = 0.4$, $\lambda_7 = \lambda_{14} = 0.3$, all (residual) variances are set to one, and the regression parameter (β) is set to either 0.1 or 0.3.

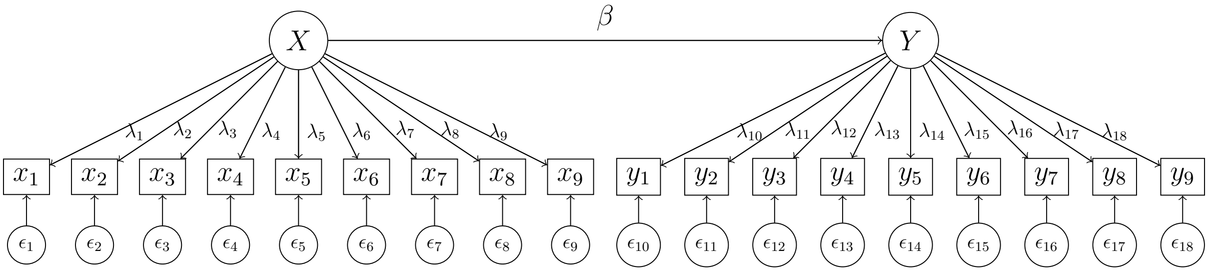

Figure 4. A simple model where the latent variable Y is regressed on the latent variable X. Each latent variable is measured by nine observed indicators. The population values are as follows: $\lambda_1 = \lambda_{10} = 1$, $\lambda_2 = \lambda_{11} = 0.8$, $\lambda_3 = \lambda_{12} = 0.7$, $\lambda_4 = \lambda_{13} = 0.6$, $\lambda_5 = \lambda_{14} = 0.5$, $\lambda_6 = \lambda_{15} = 0.4$, $\lambda_7 = \lambda_{16} = 0.3$, $\lambda_8 = \lambda_{17} = 0.3$, $\lambda_9 = \lambda_{18} = 0.2$, all (residual) variances are set to one, and the regression parameter (β) is set to either 0.1 or 0.3.

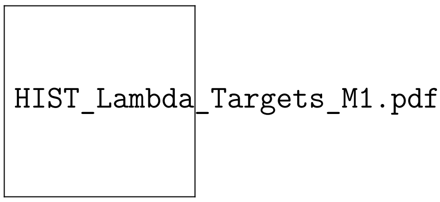

Figure 5. The distribution of $\lambda > 0$ for converged cases and for a population model where $\beta = 0.1$ and N = 30 (see Table 4).

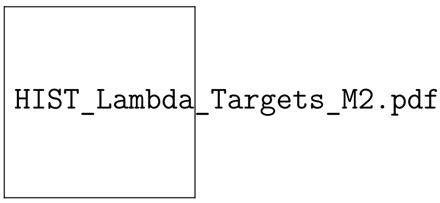

Figure 6. The distribution of $\lambda > 0$ for converged cases and for a population model where $\beta = 0.3$ and N = 30 (see Table 4).

Appendix A

Example model-based target: R code

```
## ------------------------------
##
## Script name: Rcode_example_ML_MB_Lambda.R
##
## Purpose:     Performance of adjusted covariance matrix using different model
##              based target matrices: (1    lambda)*S + lambda*T.
##
## Authors:     De Jonckere J. and Rosseel Y.
##              Department of Data Analysis, Ghent University
##
## ------------------------------
## ------------------------------
source("Get_Options_TsemStar.R")

# population model
pop.model <- '
# factor loadings
Y =~ 1*y1 + 0.8*y2 + 0.6*y3
X =~ 1*x1 + 0.8*x2 + 0.6*x3

# regression part
Y ~ 0.25*X
'

# model to be fitted
model <- model <- '
    Y =~ y1 + y2 + y3
    X =~ x1 + x2 + x3

    Y ~ X
'


set.seed(12)
Data <- simulateData(pop.model, sample.nobs = 20L)
fit.dummy <- sem(model, data = Data, do.fit = FALSE)

## ------------------------------------------------------------------------ ##
##                      Option 0: standard ML                               ##
## ------------------------------------------------------------------------ ##
fit.ML <- sem(model, data = Data, estimator = "ML", optim.attempts = 1L,
              verbose = TRUE)
summary(fit)

covmatrix <- lavInspect(fit.ML, 'samplestats', add.class = FALSE)$cov
covmatrix


## ------------------------------------------------------------------------ ##
##              Option 1: user-defined correlation matrix                   ##
```

```
## ---------------------------------------------------------------------------- ##
user.cor <- matrix(c(1,0.4,0.4,1), nrow = 2, ncol = 2)
cov.MB1  <- lav_samplestats_modify_cov_ud(lavsamplestats = fit.dummy@SampleStats,
                                          lavmodel = fit.dummy@Model,
                                          lambda = 0.2,
                                          COR.M = user.cor,
                                          reliability = 0.8)
fit.covDP1 <- sem(model, sample.cov = cov.MB1[[1]][[1]]$cov, sample.nobs = 20,
                  sample.cov.rescale = FALSE)
summary(fit.covDP1)

cov.MB1[[1]][[1]]$cov

## ---------------------------------------------------------------------------- ##
##                      Option 2: single correlation                            ##
## ---------------------------------------------------------------------------- ##
cov.MB2 <- lav_samplestats_modify_cov(lavsamplestats = fit.dummy@SampleStats,
                                      lavmodel = fit.dummy@Model,
                                      lambda = 0.2,
                                      cor = 0.2,
                                      reliability = 0.8)

fit.covDP2 <- sem(model, sample.cov = cov.MB2[[1]][[1]]$cov, sample.nobs = 20,
                  sample.cov.rescale = FALSE)
summary(fit.covDP2)

cov.MB2[[1]][[1]]$cov

## ---------------------------------------------------------------------------- ##
##                   Option 3: zero latent correlation                          ##
## ---------------------------------------------------------------------------- ##
# construct the model-based variance-covariance matrix
cov.MB3 <- lav_samplestats_modify_cov(lavsamplestats = fit.dummy@SampleStats,
                                      lavmodel = fit.dummy@Model,
                                      lambda = 0.2,
                                      cor = 0,
                                      reliability = 0.8)

fit.covDP3 <- sem(model, sample.cov = cov.MB3[[1]][[1]]$cov, sample.nobs = 20,
                  sample.cov.rescale = FALSE)
summary(fit.covDP3)

cov.MB3[[1]][[1]]$cov

# try another value for lambda that is lower than 0.5.
#cov.DP.lower <- lav_samplestats_modify_cov(lavsamplestats = fit.dummy@SampleStats,
#                                           lavmodel = fit.dummy@Model,
#                                           lambda = 0.1,
#                                           cor = 0,
#                                           reliability = 0.8)

#fit.covDP.lower <- sem(model, sample.cov = cov.DP.lower[[1]][[1]]$cov,
#                       sample.nobs = 20, sample.cov.rescale = FALSE)
```

```
#summary(fit.covDP.lower)


## ------------------------------------------------------------------------------ ##
##                    Option 4: cov elements Sa = cov S                          ##
## ------------------------------------------------------------------------------ ##
# try model-based with covariances Sa = covariances S
cov.MB4 <- lav_samplestats_modify_cov2(lavsamplestats = fit.dummy@SampleStats,
                                        lavmodel = fit.dummy@Model,
                                        lambda = 0.25,
                                        cor = 0,
                                        reliability = 0.8)

fit.covDP4 <- sem(model, sample.cov = cov.MB4[[1]]$cov, sample.nobs = 20,
                  sample.cov.rescale = FALSE)
summary(fit.covDP4)
cov.MB4[[1]]$cov
```

Appendix B

Example model-based target: Output standard ML fit

```
> summary(fit.2)
lavaan 0.6-10 did NOT end normally after 1723 iterations
** WARNING ** Estimates below are most likely unreliable

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        15
  Number of equality constraints                     2

  Number of observations                            20

Model Test User Model:

  Test statistic                                    NA
  Degrees of freedom                                NA

Parameter Estimates:

  Standard errors                             Standard
  Information                                 Expected
  Information saturated (h1) model          Structured

Latent Variables:
                   Estimate   Std.Err  z-value  P(>|z|)
  Y =~
    y1                0.915       NA
    y2                1.369       NA
    y3                0.716       NA
  X =~
    x1                0.000       NA
    x2                3.000       NA
    x3               -0.000       NA

Regressions:
                   Estimate   Std.Err  z-value  P(>|z|)
  Y ~
    X                 0.000       NA

Variances:
                   Estimate   Std.Err  z-value  P(>|z|)
   .y1                1.112       NA
   .y2                0.710       NA
   .y3                0.387       NA
   .x1                1.265       NA
   .x2           -53684.894       NA
   .x3                0.999       NA
   .Y                 0.644       NA
    X              5965.209       NA
}
```